

Chapter 45: The Matrix of a Linear Transformation

Goal: If $T : V \rightarrow W$ is a linear transformation of an n -dimensional vector space V to an m -dimensional vector space W , and if we choose bases for V and W , then we can associate a unique $m \times n$ matrix A with T that will enable us to find $T(\mathbf{x})$ for $\mathbf{x}$ in V by doing matrix multiplication.

Recall: The **coordinate vector** of $\mathbf{v}$ with respect to B is

$$C_B(\mathbf{v}) = C_B(a_1\mathbf{b}_1 + a_2\mathbf{b}_2 + \dots + a_n\mathbf{b}_n) = a_1\mathbf{e}_1 + a_2\mathbf{e}_2 + \dots + a_n\mathbf{e}_n =$$

Note: $C_B(\mathbf{b}_i) = \mathbf{e}_i$, where $\mathbf{e}_i$ is the i th standard basis vector for $\mathbb{R}^n$. This is true for ANY V and B .

Example: Let $B = (x+1, 3)$ be an ordered basis for P_1 . Find $C_B(\mathbf{v})$ for $\mathbf{v} = 2x - 4$.

Theorem: If V has dimension n and $B = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ is any ordered basis of V , the coordinate transformation $C_B : V \rightarrow \mathbb{R}^n$ is an isomorphism. In fact, $C_B^{-1} : \mathbb{R}^n \rightarrow V$ is given by

$$C_B^{-1} \left(\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \right) = \quad \text{for all } \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix} \text{ in } \mathbb{R}^n.$$

Example: Let $B = (x+1, 3)$ be an ordered basis for P_1 . What's $C_B^{-1} \left(\begin{bmatrix} 4 \\ -5 \end{bmatrix} \right)$?

Matrix of a Linear Transformation

Let V, W be vector spaces with $\dim(V) = n$, $\dim(W) = m$

If we want to build a matrix A that does the same thing as a linear transformation $T : V \rightarrow W$, using ordered bases $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ and $D = (\mathbf{d}_1, \mathbf{d}_2, \dots, \mathbf{d}_m)$ for V and W , what would we have to do?

To compute $T(\mathbf{v})$ using the matrix A :

1. Build the _____ of $\mathbf{v}$ with respect to B , $C_B(\mathbf{v})$.
2. Multiply $C_B(\mathbf{v})$ by A to make $AC_B(\mathbf{v}) = T_A(C_B(\mathbf{v}))$.
3. “Translate” $AC_B(\mathbf{v})$ back to W : _____.

The following diagram illustrates the relationship between T and A :

If we compute $\mathbf{b}_j$ using both sides, we obtain the following information about A :

Theorem: Let $T : V \rightarrow W$ be a linear transformation of an n -dimensional vector space V into an m -dimensional vector space W , and let $B = (\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n)$ and $D = (\mathbf{d}_1, \dots, \mathbf{d}_m)$ be ordered bases for V and W , respectively. Then the $m \times n$ matrix A whose j th column is $C_D[T(\mathbf{b}_j)]$ has the following property:

$$C_D[T(\mathbf{v})] = AC_B(\mathbf{v}) \text{ for every } \mathbf{v} \text{ in } V.$$

Moreover, A is the only matrix with this property.

The matrix A is called the **matrix of T corresponding to the ordered bases B and D** , and we use the following notation:

$$M_{DB}(T) = [C_D[T(\mathbf{b}_1)] \quad C_D[T(\mathbf{b}_2)] \quad \cdots \quad C_D[T(\mathbf{b}_n)]]$$

Note: We'll denote this matrix as $M_{DB}(T)$ (instead of A) from now on. Your book denotes it as $[T]_{DB}$.

Example: Let $L : P_2 \rightarrow P_1$ be defined by $L(p) = p'$ and consider the ordered bases $B = (x^2, x, 1)$ and $D = (x, 1)$ for P_2 and P_1 , respectively.

- (a) Find the matrix $M_{DB}(L)$ of L corresponding to B and D .
- (b) For $p = x^2 - 5x + 7$, compute $L(p)$ both directly and by using $M_{DB}(L)$.

Example: Let L be the same linear function as above, but use ordered bases $\tilde{B} = (1, x, x^2)$ and $\tilde{D} = (x + 1, x - 1)$ for P_2 and P_1 respectively.

- (a) Find the matrix $M_{\tilde{D}\tilde{B}}(L)$ of L corresponding to $\tilde{B}$ and $\tilde{D}$.
- (b) If $p = x^2 - 5x + 7$, compute $L(p)$ by using $M_{\tilde{D}\tilde{B}}(L)$.

We will now see that composition of linear transformations is equivalent to matrix multiplication:

Theorem: Let $T : V \rightarrow W$, and $S : W \rightarrow U$ be linear transformations and let B , D , and E be finite ordered bases of V , W , and U , respectively. Then

$$M_{EB}(ST) = \underline{\hspace{2cm}}$$

Properties of a transformation from properties of its matrix

Theorem: Let $T : V \rightarrow W$ be a linear transformation, with ordered bases B and D of V and W , respectively. Then:

1. A vector $\mathbf{v}$ in V is in $\ker T$ if and only if $C_B(\mathbf{v})$ is in the null space of $M_{DB}(T)$.
2. The dimension of the kernel of T is equal to dimension of the null space of $M_{DB}(T)$.
3. A vector $\mathbf{w}$ in W is in $\text{im } T$ if and only if $C_D(\mathbf{w})$ is in the column space of $M_{DB}(T)$.
4. The dimension of the image of T is equal to the dimension of the column space of $M_{DB}(T)$.

Proof: (Pick 1 or 3):

Theorem: Let $T : V \rightarrow W$ be a linear transformation, where $\dim V = \dim W = n$. The following are equivalent.

1. T is an isomorphism.
2. For every choice of ordered bases B and D of V and W , $M_{DB}(T)$ is invertible.
3. For at least one choice of ordered bases B and D of V and W , $M_{DB}(T)$ is invertible.

When this is the case, $[M_{DB}(T)]^{-1} = M_{BD}(T^{-1})$.

Example: $L : P_2 \rightarrow P_1$ be defined by $L(p) = p'$. Compute $\dim \operatorname{im} L$.

Example: Let $T : P_1 \rightarrow P_1$ by $T(p) = xp'$. Is T an isomorphism?

Chapter 46: Similarity of Linear Operators

Recall: A linear transformation from V to itself i.e. $T : V \rightarrow V$ is called a **linear operator**.

In this case, we can choose the SAME ordered basis B for V and the target space (also V). The matrix of T in this case has a special name/notation:

Definition: If $T : V \rightarrow V$ is an operator on a vector space V , and if B is an ordered basis of V , define $M_B(T) = M_{BB}(T)$. Some books call this the **B -matrix** of T . Your book writes $M_B(T)$ as $[T]_B$.

Question: For a linear operator $T : V \rightarrow V$, how does $M_B(T)$ change depending on the choice of basis B ?

Starting question: How do the coordinates of a vector change when we change bases?

Recall: For a vector space V with ordered bases $B = (\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ and D , the transition matrix $P_{D \leftarrow B} = [C_D(\mathbf{v}_1) \ C_D(\mathbf{v}_2) \ \cdots \ C_D(\mathbf{v}_n)]$ gives us

$$P_{D \leftarrow B} C_B(\mathbf{v}) =$$

Example: Let $B = (1, x)$ and $D = (x + 1, x - 1)$ be two ordered bases for P_1 , and build $P_{D \leftarrow B}$.

Theorem: For ordered bases B and D for a vector space V , $P_{D \leftarrow B} = M_{DB}(T)$ for $T = \underline{\hspace{1cm}}$

Set-up: Suppose we have a linear operator $T : V \rightarrow V$, and two ordered bases B and D for V . How do $M_B(T)$ and $M_D(T)$ relate?

Theorem: (Similarity Theorem) Let B and D be two ordered bases of a finite dimensional vector space V . If $T : V \rightarrow V$ is any linear operator, the matrices $M_B(T)$ and $M_D(T)$ of T with respect to these bases are similar. More precisely,

$$M_B(T) = P^{-1}M_D(T)P =$$

where $P = P_{D \leftarrow B}$ is the change matrix from B to D .

The diagram below illustrates the theorem:

Example: Let $L : P_1 \rightarrow P_1$ by $L(ax + b) = bx + (2a + b)$. Using ordered bases $B = (1, x)$ and $D = (x + 1, x - 1)$:

- (a) Build $M_D(L)$.
- (b) Use the Similarity Theorem to compute $M_B(L)$.

Example: Let $L : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be defined by

$$L \left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \right) = \begin{bmatrix} 2u_1 - u_3 \\ u_1 + u_2 - u_3 \\ u_3 \end{bmatrix}.$$

Let D be the standard basis for $\mathbb{R}^3$. Then the D -Matrix of L is

$$M_D(L) = \begin{bmatrix} 2 & 0 & -1 \\ 1 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix}.$$

Now consider the ordered basis $B = \left(\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right)$ for $\mathbb{R}^3$. What is $M_B(L)$, given that the transition matrix P from B to D has inverse

$$P^{-1} = \begin{bmatrix} 0 & 0 & 1 \\ -1 & 1 & 1 \\ 1 & 0 & -1 \end{bmatrix}?$$

This raises the question: Given $T : V \rightarrow V$, can we always find a basis B such that the B -matrix of T is diagonal? If not, when can we find such a basis?

Eigenvalues of Linear Operators

Definition: A real number λ is called an **eigenvalue** of an operator $T : V \rightarrow V$ if $T(\mathbf{v}) = \lambda \mathbf{v}$ for some nonzero vector $\mathbf{v}$ in V . In this case, $\mathbf{v}$ is called an **eigenvector** of T corresponding to λ .

Example: Consider $T : P_2 \rightarrow P_2$ by $T(p(x)) = xp'(x)$. Show that x^2 is an eigenvector of T . What's the eigenvalue?

The subspace $E_\lambda(T) = \{\mathbf{v} \text{ in } V \mid T(\mathbf{v}) = \lambda \mathbf{v}\}$ is called the **eigenspace** of T corresponding to λ .

Remark: If A is an $n \times n$ matrix, then λ is an eigenvalue of the matrix operator $T_A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ if and only if λ is an eigenvalue of the matrix A , and the eigenspaces are the same.

How do we find eigenvalues/eigenvectors of a general linear operator?

Choose an ordered basis B for V , and consider $A = M_B(T)$. Suppose $T(\mathbf{v}) = \lambda \mathbf{v}$ for some nonzero vector $\mathbf{v}$.

What is $AC_B(\mathbf{v})$?

Theorem: Suppose $T : V \rightarrow V$ is a linear operator where $\dim V = n$, λ is a number, and $\mathbf{v}$ is a vector in V . Then the following are equivalent:

1. T has eigenvalue/eigenvector pair $\lambda, \mathbf{v}$.
2. For every choice of ordered basis B for V , the matrix $M_B(T)$ has eigenvalue/eigenvector pair $\lambda, C_B(\mathbf{v})$.
3. For at least one choice of ordered basis B for V , the matrix $M_B(T)$ has eigenvalue/eigenvector pair $\lambda, C_B(\mathbf{v})$.

This theorem says: To find the eigenvalues/eigenvectors for a general linear operator $T : V \rightarrow V$:

1. Pick _____ ordered basis B for V .
2. Build the matrix _____.
3. Compute the eigenvalues and eigenvectors for _____.
4. T has the _____ eigenvalues. To get the eigenvectors, “translate” back to V from $\mathbb{R}^n$ using C_B^{-1} .

Example: Let $L : P_1 \rightarrow P_1$ by $L(ax + b) = bx + (2a + b)$. We computed $M_B(L)$ and $M_D(L)$ for both ordered bases $B = (1, x)$ and $D = (x + 1, x - 1)$:

$$M_B(L) = \begin{bmatrix} 1 & 2 \\ 1 & 0 \end{bmatrix} \quad M_D(L) = \begin{bmatrix} 2 & 0 \\ -1 & -1 \end{bmatrix}$$

Find the eigenvalues and corresponding eigenspaces for each matrix, then find the eigenvalues and eigenspaces for L .

(more room for example)

Example: Let $T : P_2 \rightarrow P_2$ be a linear operator defined by

$$T(ax^2 + bx + c) = -bx - 2c.$$

What are the eigenvalues of T ?

Diagonalizability

Recall: An $n \times n$ matrix is diagonalizable if and only if $\mathbb{R}^n$ has a basis $\{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ consisting of eigenvectors of A .

Definition: A linear operator T on a finite dimensional space V is called **diagonalizable** if there is an ordered basis B of V such that $M_B(T)$ is diagonal.

Theorem: Let $T : V \rightarrow V$ be a linear operator on a finite dimensional space V . Then the following conditions are equivalent.

1. V has a _____ consisting of eigenvectors of T .
2. There exists a basis B of V such that $M_B(T)$ is diagonal.

Definition: A linear operator T on a finite dimensional space V is called _____ if V has a basis consisting of eigenvectors of T .

Example: Let $L : P_1 \rightarrow P_1$ by $L(ax + b) = bx + (2a + b)$. Find a basis U of P_1 such that $M_U(L)$ is diagonal.

Corollary: Let $T : V \rightarrow V$ be a linear operator on a finite dimensional space V . T is diagonalizable if and only if a matrix $M_B(T)$ representing T is diagonalizable.

Example: Let $T : P_2 \rightarrow P_2$ be a linear operator defined by $T(ax^2 + bx + c) = -bx - 2c$. Is T diagonalizable?